Bit-level reproduction of a driven quasiparticle-kinetics benchmark and the sampling-convention sensitivity of gap-suppression figures

[TODO: author list and affiliations]

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Numerical figures in nonequilibrium superconductivity are routinely used as benchmarks for new kinetic codes, yet the discretization conventions behind them are rarely reported at the level a reproduction requires. We introduce a component-substitution validation method — a *reproduction ladder* — and apply it to Figure 6 of Fischer and Catelani [Phys. Rev. Applied **XX** (2023)], which shows the suppression of the superconducting gap under a sub-gap photon drive. Starting from the authors' own program, we first reproduce their plotted ordinate exactly (to the last floating-point bit) with an independent numerical port, then replace one scientific component at a time — observable, parameters, energy grid, photon operator, electron–phonon operators, phonon balance, and finally the nonlinear solver — with the corresponding public component of our open kinetic solver QPSIM, comparing every collision channel on identical frozen states before any re-solve. All collision physics agrees to within a few parts in 10^{15}. The entire historical discrepancy between the codes — 33–39% on the rising branch of the published figure — is attributed to three labeled numerical choices, dominated by a half-bin difference in where the occupation function is sampled on the energy grid: the same solved state reads 0.1454 under the authors' left-edge convention and 0.0505 under the cell-center convention at the anchor point, versus the published 0.1209. Two smaller policy differences — the one-bin labeling of recombination-phonon frequencies and an analytic (Kaplan) treatment of the near-threshold pair-breaking quadrature that the original midpoint sum overestimates by up to 27% per bin — are isolated with dedicated controls. Grid refinement to $\mathrm{d}E = 0.25\,\mu\mathrm{eV}$ shows the center-carrier reading already converged at the published resolution while the left-edge reading still moves by $\sim 9\%$ per halving — the published observable carries a discretization offset comparable to the inter-code discrepancy itself. We argue that figures constructed from ratios of nearly equal gap shifts amplify such sampling offsets by more than an order of magnitude, and recommend that published kinetic benchmarks state their sampling convention and demonstrate grid convergence of the plotted observable.

I. INTRODUCTION

Nonequilibrium quasiparticle (QP) kinetics in superconducting devices — readout-induced QP generation, phonon trapping, gap suppression under microwave drive — is increasingly modeled with dedicated kinetic solvers. When a new solver is introduced, the community reasonably asks that it reproduce established results. This paper is about what "reproduce" should mean, and about a specific, quantitatively large pitfall we found when we held ourselves to a strict standard.

QPSIM is an open kinetic solver for driven superconducting QP and phonon distributions [TODO: one-paragraph software description: energy grid, collision operators, diffusion tier, junction layer; cite repository/release]. As part of its validation we targeted Figure 6 of Fischer and Catelani [1] (henceforth F&C), which plots the normalized gap suppression of a superconductor driven by a sub-gap photon mode of energy $\omega_0 = 20\,\mu\mathrm{eV}$ against an effective drive temperature T^*/Δ, for bath temperatures $T_B \in \{0.10, 0.15, 0.20\}$ K. Our initial comparison showed QPSIM lower than the published solid curves by 33–39% on the rising branch — far outside any acceptable tolerance, and, importantly, a discrepancy whose cause could not be localized by staring at final curves.

The authors kindly provided the program used to produce the figure [TODO: confirm acknowledgment wording with F&C]. This enabled a validation design in which the question "why do the codes disagree?" is answered constructively: by transforming one program into the other, one scientific component at a time, with machine-checked evidence at every step.

Three results came out of this campaign. *First*, a bit-level reproduction: an independent numerical port reproduces the authors' plotted ordinate at an anchored point exactly, and QPSIM's public observable, parameters, grid, photon operator, electron–phonon operators, and phonon balance each reproduce the corresponding author computation to within a few parts in 10^{15} on identical states. *Second*, an attribution: the entire historical discrepancy reduces to three labeled numerical choices, none of which is a defect in the collision physics of either code. *Third*, a warning of general interest: the plotted observable is so sensitive to the energy-grid sampling convention that a half-bin ($0.5\,\mu\mathrm{eV}$) offset in where the occupation is read moves the anchored ordinate by nearly a factor of two. Figures of this type — ratios of nearly equal gap integrals over exponentially falling distributions weighted by the BCS gap-edge singularity — amplify discretization offsets far beyond the naive $\mathcal{O}(\mathrm{d}E)$ expectation.

II. THE REPRODUCTION LADDER

The method separates three questions that figure-level comparisons conflate: (i) what did the authors' program actually compute; (ii) can a small independent implemen-

TABLE I. The Figure-6 reproduction ladder. Every completed stage is bound by a committed, hash-anchored score and receipt; "exact" means bit-for-bit.

Stage	Changed component	Outcome at the anchored point
A0/A1	— (authentication)	author program and output authenticated; one exact full-resolution point replayed
C0	— (clean-room port)	author ordinate 0.12090908988993258 reproduced exactly; final state to 4.0×10^{-16}
C1	observable	bit-exact
C2	parameters	bit-preserving plumbing; declared constant updates
C3	grid sampling	ordinal embedding exact; conventions labeled (coherence, pair labels, cell density)
C4	photon operator	net agrees to 2.0×10^{-15}
C5	QP–phonon operator	nets agree to 5.7×10^{-16}
C6	phonon balance	roundoff except the Kaplan quadrature policy (Sec. III B 2)
C7	nonlinear solver	converges in the same 4 iterations; residual assembly matches C6 to 1.3×10^{-13}
Q0	(endpoint)	full-curve sweep under both sampling conventions (Sec. III F)

tation reproduce it; (iii) at which first component does substituting the new code change the result?

Each *stage* of the ladder declares exactly one changed scientific component out of nine (observable, parameters, grid sampling, photon operator, QP–phonon operator, phonon balance, nonlinear solver, I/O capture, physical model). A machine-readable contract (`reproduction-ladder.json`) rejects any stage that changes zero or more than one component, and requires completed stages to bind implementation and result evidence by SHA-256. Collision operators are compared on *identical frozen states* before any nonlinear re-solve is permitted, so that the first differing equation is visible instead of being laundered through solver feedback. Every stage's raw arrays are retained in an immutable bundle; a committed *receipt* binds the bundle digest and the complete checked-score bytes; and an *independent verifier* — which does not import the producer code or the substituted public operators — re-derives every kernel, contraction, and reduction with fixed-order compensated arithmetic and predeclared IEEE-754 error bounds, so acceptance is host-independent.

Table I summarizes the stages. The anchored comparison point is $T_B = 0.20\,\mathrm{K}$ at the authors' sweep index 49 ($T^*/\Delta = 0.33990789737294363$), the retained point nearest the digitized-figure coordinate.

III. RESULTS

A. The collision physics agrees to roundoff

On the frozen anchored state, QPSIM's public operators reproduce the author channels with the following symmetric relative L^1 differences: photon channel 2.0×10^{-15} (the only structural difference being two terminal-bin transitions of magnitude $\sim 3 \times 10^{-35}\,\mathrm{s}^{-1}$ that the author residual omitted); QP-side scattering-plus-pair net 5.7×10^{-16} (after accounting for a bookkeeping difference in which the author source-order buckets carry the same Pauli cross-term in both gain and loss); phonon-side scattering 1.3×10^{-15}; phonon-side pair channel 2.4×10^{-16} with the quadrature policy of Sec. III B 2 disabled; and phonon escape at the 10^{-16} level. Detailed balance of the substituted phonon-side operators at a native thermal control holds to below 6.3×10^{-16} of channel turnover. In short: neither code's collision physics is wrong, and no amount of further kernel debugging could have closed the figure-level gap.

B. Three labeled numerical choices

1. *Pair-frequency labeling (± one bin)*

The author grid stores quantities at cell *left edges*; QPSIM stores them at cell *centers*. A recombination event between cells i and j therefore emits a phonon labeled $2\Delta + (i + j)h$ in the author code and $2\Delta + (i + j + 1)h$ in QPSIM ($h = 1\,\mu\mathrm{eV}$). A same-seed re-solve isolating only this label moves the anchored ordinate from 0.1209 to 0.1459; the completed C7 re-solve with all components substituted lands at 0.1454 under the authors' observable convention.

2. *Near-threshold pair-breaking quadrature*

Both codes evaluate the phonon-side pair-breaking rate by a per-bin sum over QP pairs whose integrand carries the BCS inverse-square-root singularity at both endpoints. The author code uses the plain per-bin (midpoint-type) sum. QPSIM rescales each complete-pair-interval phonon bin to the analytic Kaplan total $S_+(\omega/\Delta)/(\pi\,\tau_0^{\mathrm{PB}})$ with $S_+(x) = x\,E(1 - 4/x^2)$. At the anchored state this correction touches 929 phonon bins with factors between 0.786 and 1 — i.e., the plain quadrature *overestimates* the closest-to-threshold bins by up to 27% — and moves the frozen pair-channel net by 9.2×10^{-3} (symmetric relative L^1) while shifting the frozen phonon residual by 2.1×10^{-6}, versus 2.2×10^{-10} for a same-kernel correction-off control. Its effect on the re-solved ordinate is small and downward ($0.1459 \to 0.1454$). We regard the analytic treatment as an improvement, but it is a real, labeled departure from what the published figure computed.

3. The sampling convention of the observable

The dominant effect is interpretive rather than dynamical. The plotted quantity is

$$y = \frac{\Delta[f_{\rm driven}] - \Delta[f_{\rm th}]}{\Delta_0 - \Delta[f_{\rm th}]}, \qquad \Delta[f] = \Delta_0\, e^{-I[f]}, \qquad (1)$$

with $I[f] \propto \int \mathrm{d}E\, \rho(E)\, f(E)/E$ evaluated by the authors' piecewise-linear rule on the discrete grid. At $T_B = 0.2\,\mathrm{K}$ the thermal scale is $k_B T_B \approx 17\,\mu\mathrm{eV}$: the occupation falls by e^{-1} every 17 bins, and the integrand is further weighted by the gap-edge singularity of $\rho(E)$. Claiming that a stored value f_k "lives" at the cell center rather than the left edge shifts the entire distribution by $0.5\,\mu\mathrm{eV}$ relative to the weight, changing the driven and thermal integrals by +11.1% and +3.0% respectively at the frozen anchored state. The ratio construction then amplifies this into the plotted number. Reading the *same* C7 solved state:

Convention (state reading / thermal reference)	y at anchor
authors / authors (as published)	0.1454
centers / authors (mixed diagnostic)	0.0777
centers / centers (self-consistent QPSIM)	0.0505
published author value (their solve, their convention)	0.1209
historical QPSIM comparison value	0.0897

The historical "QPSIM is 33–39% low" comparison was, in effect, a center-convention number plotted against a left-edge-convention curve. Under the authors' own convention, QPSIM's observable reproduces their anchored ordinate *exactly* at their solved state. Both conventions are legitimate discretizations of the same continuum integral and converge to a common answer as $\mathrm{d}E \to 0$ (Sec. III F); the error is created only when a state produced under one convention is read under the other.

C. The solver is interchangeable

With all operators substituted, replacing the authors' explicit-inverse, undamped Newton driver by QPSIM's public coupled Newton solver (analytic cross-Jacobian, line search, scale-invariant stopping) changes nothing material: seeded from the authors' captured per-point thermal initial state (their program constructs each sweep point independently from a thermal initializer), it converges at the minimal public iteration cap of 4 — the authors' count — with L^1 residual-to-turnover ratios of 3.7×10^{-14} (QP equation) and 5.8×10^{-12} (phonon equation) at the returned root, and its residual assembly evaluated at the frozen parent state matches the accepted ladder residuals to 1.3×10^{-13}.

D. Closing the historical comparison

The historical QPSIM comparison value (0.0897 at the anchor) originated from a different *physical model*: QPSIM's self-consistent moving-gap solve, in which the driven gap is the converged fixed point of $\Delta = \Delta_0 e^{-I[f_\Delta]}$ rather than the authors' fixed-gap kinetics with a post-processed gap integral. With every leg now measured, the historical discrepancy decomposes completely at the anchor: the authors' value 0.1209 maps to ≈ 0.048 under the center-consistent reading of the fixed-gap model (the sampling convention, −0.095, partially offset by the pair-label and Kaplan policies, +0.02 net), and the self-consistent gap feedback then raises it to ≈ 0.091 (+0.04; promoted regression row at $T^*/\Delta = 0.3426$). The partial cancellation of the last two legs is why the historical curves looked "only" 30–40% apart rather than the factor ~ 2.4 that the sampling convention alone produces — and why staring at final curves could never have attributed the difference.

E. What the certificates refuse at 0.10 K

An instructive side finding: at $T_B = 0.10$ K QPSIM's certificate-mode solver *refuses* to return, at any drive in the sweep and at any certificate threshold. In double precision the attainable phonon-equation balance floors at $\approx 2 \times 10^{-7}$ of turnover (the pair-channel normalization collapses as $e^{-2\Delta/k_B T_B}$), and the backtracking line search dies at the residual noise floor before any certificate is even evaluated. We then attempted the 0.10 K curve in the solver's legacy absolute-residual mode near the measured floor, *instrumenting every returned point with measured residual-to-turnover ratios* — and the instrumentation condemned the curve: weak-drive points returned their seed state unchanged (the classic absolute-tolerance stale exit), and strong-drive points returned unphysical states with measured imbalances up to 7×10^{-2} of turnover and absurd ordinates. Isolated cold-seed spot checks at intermediate drive do converge with excellent measured quality (QP $\sim 10^{-11}$, phonon $\sim 2 \times 10^{-7}$), but a trustworthy full 0.10 K curve is not obtainable from this solver configuration in double precision, and we exclude that temperature from the curve figure rather than plot uncertified points. Two lessons travel beyond this code: "the solver returned" is not a statement of accuracy in the cold regime — the authors' fixed ten-step Newton driver attaches no quality statement at all at 0.10 K — and per-point measured balance ratios are cheap and catch both failure modes. [TODO: future work: compensated/extended-precision residual assembly to certify the 0.10 K curve.]

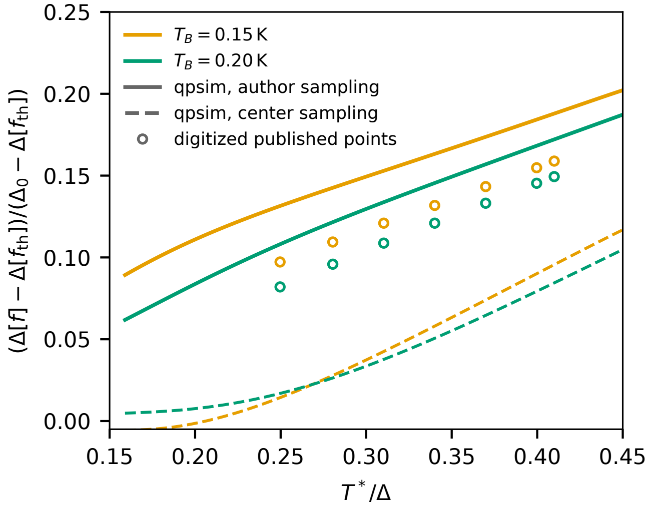

FIG. 1. The author-semantics QPSIM endpoint over the published window ($\mathrm{d}E = 1\,\mu$eV, certificate-mode solves; $T_B = 0.15$ K at a documented one-decade-relaxed certificate at its float64 floor). Solid: the solved occupation read under the authors' left-edge convention; dashed: the same states read at cell centers; open circles: the published numerical curves digitized from the arXiv raster (error bars smaller than the markers). The published points lie *between* the two readings of one and the same solution family; $T_B = 0.10$ K is excluded per Sec. III E. Diagnostic artifact: `fig6-Q0-sweep-de1-v1.json` (+`-t015-v1`).

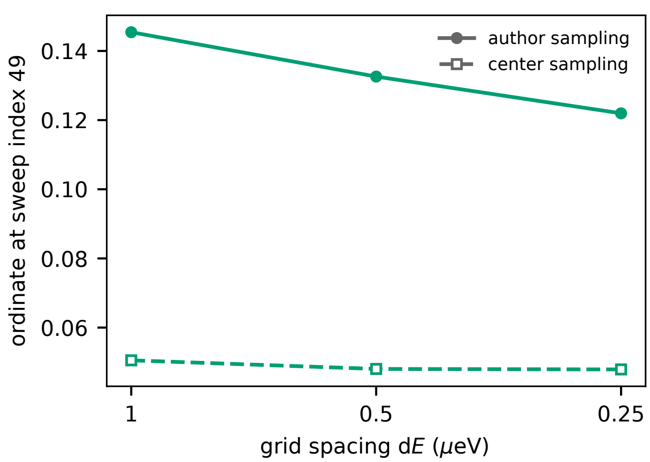

FIG. 2. Grid refinement of the anchored-point ordinate ($T_B = 0.20$ K, sweep index 49) under both readings of the re-solved root. The center-carrier reading is grid-converged at the published resolution; the left-edge reading is not.

F. Full-curve endpoint and grid refinement

Grid refinement at the anchored point ($T_B = 0.20$ K, sweep index 49) gives, for the re-solved root read under each convention:

dE (μeV)	1	0.5	1/3	0.25	0.125
author sampling	0.1454	0.1326	0.1262	0.1220	[TODO: de8]
center sampling	0.0505	0.0480	0.0477	0.0479	[TODO: de8]

The center-carrier reading is grid-converged at the published resolution (stable within ± 0.0002 from $\mathrm{d}E = 0.5\,\mu$eV on), whereas the left-edge reading of the same solved states is still moving at $\mathrm{d}E = 0.25\,\mu$eV, with successive differences decaying *sub-linearly* (apparent order ~ 0.3, consistent with an edge-cell-dominated quadrature error). Both readings are consistent discretizations of the same continuum functional and must meet as $\mathrm{d}E \to 0$; the measured sequence already shows that the published observable, at the published $1\,\mu$eV resolution and under its own sampling convention, carries a discretization offset of at least $\sim 20\%$ relative to its refined value — comparable to the entire inter-code discrepancy that motivated this study. We deliberately refrain from a three-parameter continuum extrapolation: the four points do not determine the limit robustly, and the slow left-edge decay merits its own analytic treatment of the singular gap-edge cells. [TODO: physicist review; fill the $\mathrm{d}E = 0.125\,\mu$eV column when the run completes.]

[TODO: Insert refinement figure (`fig_refinement.pdf`) and the 0.10/0.15 K refinement columns; data: `fig6-Q0-sweep-de2-v1.json`, `fig6-Q0-sweep-de4-v1.json`, `fig6-Q0-sweep-de2-t010-v1.json`.]

IV. DISCUSSION AND RECOMMENDATIONS

The campaign closes a two-code discrepancy without finding a physics bug in either code, which is precisely why we think it is worth reporting. Had we tuned kernels until the curves overlapped, we would have "fixed" correct physics. The actionable conclusions:

1. **Report the sampling convention.** For kinetic benchmarks built from exponentially falling distributions under singular spectral weights, state where on the grid the distribution is sampled and how the observable quadrature pairs with that choice. A half-bin ambiguity is invisible in a methods section and worth a factor of two in a figure of this construction.

2. **Demonstrate grid convergence of the plotted observable**, not merely of the solver residual. The residual converges long before the convention dependence of a ratio-of-small-differences observable is resolved.

3. **Ratio observables amplify.** The normalization $\Delta_0 - \Delta[f_{th}]$ is itself small; offsets that perturb driven and thermal integrals unequally are magnified by an order of magnitude or more in the plotted quantity.

4. **Near-threshold quadrature of pair-breaking deserves analytic treatment.** The per-bin error of the plain sum (up to 27% in the closest bins here) is silent in aggregate observables but is a real rate distortion at threshold.

5. **Component-substitution validation is practical.** The entire ladder — authentication, cleanroom port, seven substitutions, independent verifiers, and machine-checked provenance — was executed against a single anchored point first, which is what made each difference attributable. Curve-level comparisons alone could not have produced Table I.

REPRODUCIBILITY

Every completed stage is bound by a committed score and receipt carrying SHA-256 digests of the immutable raw-array bundles, the complete source closure (the full QPSIM tree plus all validation modules), and the runtime; the independent verifiers use fixed-order compensated summation with predeclared error bounds so their acceptance is host-independent. The authors' program is authenticated as author-supplied reproduction source [TODO: confirm redistribution / citation terms with F&C before publication]. [TODO: repository DOI / release tag].

ACKNOWLEDGMENTS

We thank M. Fischer and G. Catelani for providing the program used to produce the original figure. [TODO: confirm; funding; colleagues.]

[1] M. Fischer and G. Catelani, [TODO: full citation of the 2023 Phys. Rev. Applied paper (Fig. 6)].

[2] S. B. Kaplan, C. C. Chi, D. N. Langenberg, J. J. Chang, S. Jafarey, and D. J. Scalapino, Phys. Rev. B **14**, 4854 (1976). [TODO: qpsim software citation; digitization/raster-oracle method note.]